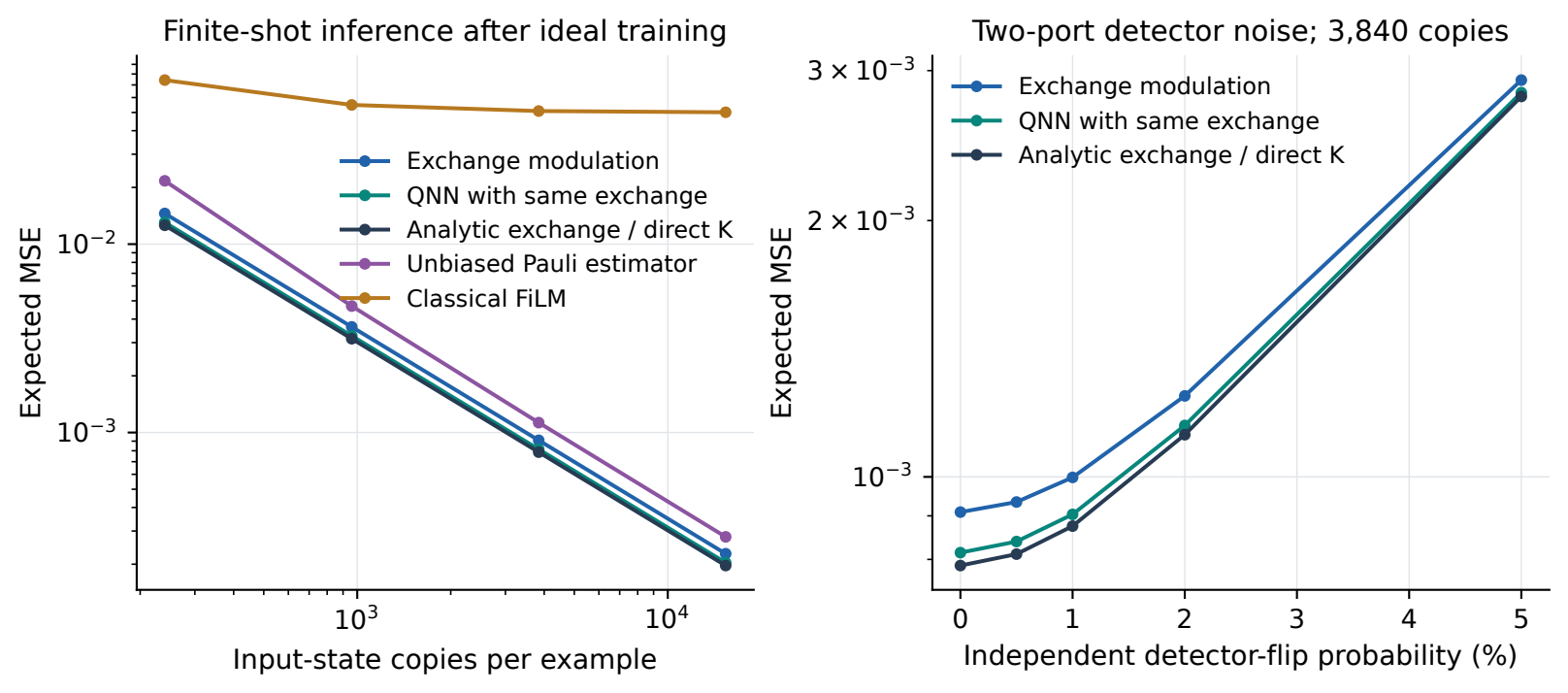
Finite-shot inference after ideal training
Expected MSE
10^{-2}
10^{-3}
10^{3}
10^{4}
Input-state copies per example
Exchange modulation
QNN with same exchange
Analytic exchange / direct K
Unbiased Pauli estimator
Classical FiLM
Two-port detector noise; 3,840 copies
Expected MSE
3×10^{-3}
2×10^{-3}
10^{-3}
0
1
2
3
4
5
Independent detector-flip probability (%)
Exchange modulation
QNN with same exchange
Analytic exchange / direct K